

AMCS2104 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions like humans. To develop an AI system, four different approaches are implemented: Acting Humanly, Thinking Humanly, Thinking Rationally, and Acting Rationally. Briefly explain the **FOUR (4)** approaches. (8 marks)
- b) Many modern technologies are related to or inspired by the idea behind the Turing Test. One example is **Meta AI**, an artificial intelligence assistant created by Meta, the company that owns Facebook, Instagram, WhatsApp, and Messenger. Meta AI is built into these apps to help users with different tasks. It can answer questions, give suggestions, help you chat, create images or animations, and assist with personalising or creating content. In simple words, Meta AI is like a smart helper inside Meta's apps that makes everyday tasks easier.
- (i) Identify how the *total Turing Test* is different from the *Turing Test*. (2 marks)
 - (ii) Identify whether Meta AI can pass the *Turing Test*. Justify your answer by focusing on the capabilities of Meta AI. (1 + 2 marks)
 - (iii) Identify whether Meta AI can pass the *total Turing Test*. Justify your answer by focusing on the capabilities of Meta AI. (1 + 2 marks)
 - (iv) Which one is more relevant to the definition of AI, the *Turing Test* or the *total Turing Test*? Justify your answer. (1 + 2 marks)
- c) Define agent in terms of AI. (2 marks)
- d) At any given time, rationality depends on four criteria to develop a rational agent. Briefly explain the **FOUR (4)** criteria. (4 marks)

[Total: 25 marks]

Question 2

A company developing smart traffic control systems aims to optimise the timing of traffic lights across a busy city to reduce congestion and improve traffic flow. Due to the dynamic nature of vehicle movements and unpredictable traffic patterns, traditional algorithms struggle to maintain efficiency during peak hours. To overcome this, the company implements an evolutionary algorithm (EA) to optimise traffic signal timings. The algorithm evolves different timing strategies and evaluates them based on average vehicle waiting time, traffic flow rate, and fuel efficiency.

- a) Analyse how an evolutionary algorithm can help improve the traffic light timing system. (2 marks)

- b) Identify and briefly explain the **FIVE (5)** phases of an evolutionary algorithm involved in optimising the traffic light timing. (5 + 10 marks)

Question 2 (Continued)

- c) Identify and briefly explain **TWO (2)** stopping criteria for the evolutionary algorithm that ensure optimal performance without excessive computation time. Justify your recommendation. (8 marks)

[Total: 25 marks]

Question 3

- a) In AI, game tree search refers to examining all possible moves and their outcomes in a game to determine the optimal decision. Based on the game tree shown in **Figure 1**, answer the following questions.

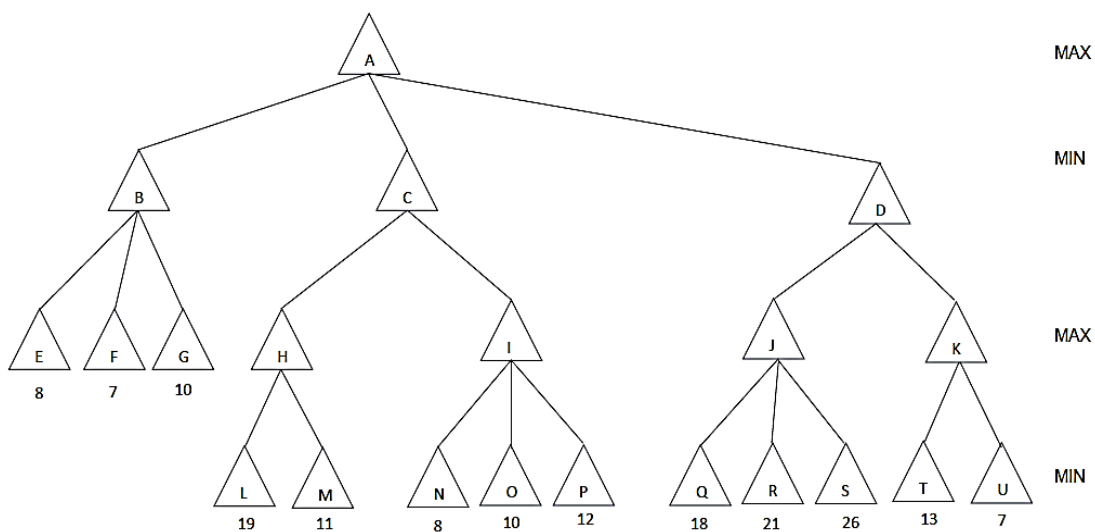


Figure 1. A partial game tree that illustrates the gameplay of a chess game

- (i) The game represented by the game tree in **Figure 1** demonstrates that the environment involves multiple agents. Identify and briefly explain the specific type of multi-agent environment depicted. (2 marks)
- (ii) Suppose the *minimax algorithm* is used to explore the game tree in search of the optimal move, compute the *minimax value* for nodes A, B, C, D, H, I, J, and K. (4 marks)

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- (iii) Identify the best move for the MAX player at node A. Justify your answer. (2 marks)
- b) *Hill climbing* is an optimisation algorithm used to identify the best possible solution for a problem. It is commonly applied in AI and machine learning for local search optimisation.
- (i) *Hill climbing* is also known as a greedy approach or greedy local search. Briefly explain why it is referred to in this way. (2 marks)
- (ii) Give **TWO (2)** strengths of *Hill Climbing*. (2 marks)
- (iii) Draw and label a state-space landscape in which its elevation corresponds to an objective function. (7 marks)

Question 3 b) (Continued)

- (iv) Hill climbing search often gets stuck at a sub-optimal solution. Identify and analyse **TWO (2)** causes for this. (2 + 4 marks)

[Total: 25 marks]

Question 4

The IT Department at TAR UMT has developed an AI-driven campus Wi-Fi optimization system designed to identify the most efficient connection path from Access Point A to Access Point I. **Figure 2** illustrates a directed network graph consisting of nine access points, where the values represent the signal transmission costs between each point. **Table 1** shows the heuristic cost of each access point relative to Access Point I, assisting the system in estimating the optimal data transmission route.

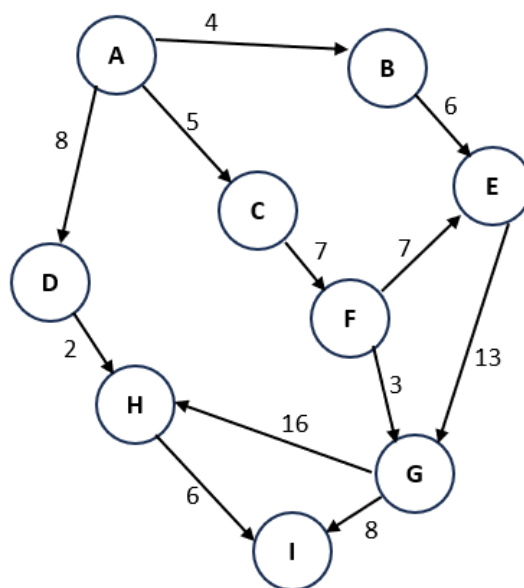
**Figure 2.** State space

Table 1: Heuristic Cost of Each Access Point

Access Point	A	B	C	D	E	F	G	H	I
Heuristic Cost	13	17	6	9	19	7	6	9	0

- a) Formulate the state space, initial state, goal state, actions, transition model, and the cost function for the problem above. (6 marks)
- b) Based on the information provided in **Figure 2** and **Table 1**, illustrate the resulting search tree for *Breadth-First Search*, *Depth-First Search*, and *A* Search* in alphabetical order. (16 marks)
- c) Identify the solution returned by *Breadth-First Search*, *Depth-First Search*, and *A* Search* in **Question 4 b)**. (3 marks)

[Total: 25 marks]